

NLP - HW2

Task 1: BERT Classification

Approach

The bert-base-uncased model processes inputs formatted as [CLS] fact stem choice [SEP]. This produces four separate sequences per question. A linear layer maps the [CLS] token embedding from the final hidden state to a single logit. Cross-entropy loss across the four logits identifies the correct choice. Training uses automatic mixed precision and early stopping based on validation loss.

Accuracies

- Zero-shot Validation: 21.60%
- Fine-tuned Test: 69.00%

Limitations and Solutions

The current implementation evaluates each sequence independently. The model does not compare the four choices during the same forward pass. Concatenating all choices into a single sequence allows the attention mechanism to compute weights across all options simultaneously.

Task 2: Generative Loss Scoring

Approach

The TransformerGPT model calculates sequence loss for text combining the fact, stem, and choice. This repeats for all four choices. The model predicts the choice with the lowest cross-entropy loss.

Accuracies

- Zero-shot Validation: 28.80%
- Zero-shot Test: 31.60%
- Fine-tuned Validation: 43.40%
- Fine-tuned Test: 49.60%

Limitations and Solutions

The loss calculation favors shorter text sequences. Shorter text yields a lower total loss regardless of semantic correctness. Normalizing the cross-entropy loss by the total token length of the choice corrects this imbalance.

Task 3: Vanilla Beam Search Generation

Approach

The model conditions on the fact and stem to generate a response using standard beam search. The generated text is compared to each of the four choices using BERTScore. The model predicts the choice with the highest BERTScore.

Accuracies

- Zero-shot Validation: 29.20%
- Zero-shot Test: 32.20%
- Fine-tuned Validation: 32.00%
- Fine-tuned Test: 31.80%

Limitations and Solutions

Standard generation lacks awareness of the specific multiple-choice options. The model generates text that is factually correct but lexically distinct from the correct choice. Integrating choice tokens into the generation constraints forces the output to align with the provided options.

Correct Examples

	Fact	Stem	Choices	Gold	Predicted	Generation	BERTScores
1	deep sea animals live deep in the ocean	Frilled sharks and angler fish live far beneath the surface of the ocean, which is why they are known as	'Deep sea animals', 'fish', 'Long Sea Fish', 'Far Sea Animals'	0	0	deep sea creatures	[0.969, 0.912, 0.918, 0.901]
2	Matter in the liquid phase has definite volume	Gas can fill any container it is given, and liquid	'is standard weight and size', 'is the opposite of variable', 'only needs a few', 'uses what it needs'	3	3	vapor	[0.706, 0.687, 0.704, 0.723]
3	migration is an instinctive behavior	When birds migrate south for the winter, they do it because	'they are genetically called to', 'their children ask for them to', 'it is important to their happiness', 'they decide to each year'	0	0	instinctive behavior is lacking	[0.865, 0.853, 0.856, 0.848]

4	the amount of daylight is greatest in the summer	If a person enjoys long, sunny days their favorite month would be	'October', 'July', 'April', 'March'	1	1	June	[0.982, 0.986, 0.981, 0.980]
5	An example of hunting is an otter cracking open clams with a rock	An animal can hunt by cracking open a	'claw', 'house', 'shell', 'bone'	2	2	shell	[0.939, 0.884, 1.0, 0.955]

Incorrect Examples

	Fact	Stem	Choices	Gold	Predicted	Generation	BERTScores
1	a compass is a kind of tool for determining direction by pointing north	If a person walks in the opposite direction of a compass arrow they are walking	'west', 'north', 'east', 'south'	3	0	in opposite direction	[0.826918489 4304169, 0.812594609 026746, 0.825403390 1487398, 0.810334262 2278557]
2	as an object moves , the kinetic energy of that object will increase	An example of lots kinetic energy would be	'Drinking a cold glass of water', 'A snail moving across the sidewalk', 'sitting without moving anywhere', 'An aircraft taking a trip'	3	2	a roller coaster	[0.736629743 57919, 0.808958863 8406231, 0.850520702 6770355, 0.765631218 1630054]
3	a single-cell organism	Which organism cannot specialize?	'mammal', 'plant', 'fish', 'protozoa	3	0	moles	[0.913872714 2285105, 0.779788883 5877863,

	cannot specialize						0.794427994 8702474, 0.836550166 8733586]
4	a greenhouse is used to protect plants from the cold	A person can grow cabbage in January with the help of what product?	['Green house', 'Parliment', 'Congress', 'White house'	0	2	greenhouse	[0.919358653 0801912, 0.906546597 3344103, 0.943188527 1092133, 0.913977989 7836539]
5	an amphibian is cold-blooded	A frog, in winter, will burrow itself into soft mud, until it freezes, then in the spring	'it melts, because it is warm-blooded', 'it unfreezes, because it is cold-blooded', 'it remains frozen until it dies', 'it dies, because it is warm-blooded'	1	3	, it dies	[[0.84325343 32758125, 0.825822633 1577407, 0.840177307 2175428, 0.901582193 6175918]

Thoughts

The model generates short phrases closely related to the core subject of the fact and stem. The generation aligns lexically with the true choice when correct. Errors occur when the generation matches a distractor lexically better than the gold choice. In example 4, the generation "greenhouse" is factually correct, but the metric scored "Congress" higher than "Green house". This indicates a failure in the scoring metric.

Task 4: Guided Beam Search Generation

Approach

The beam search algorithm incorporates a composite score for pruning decisions. The score combines generative log-likelihood with the BERTScore computed between the fact, stem, and the active generation sequence at each step.

Accuracies

- Zero-shot Validation: 29.40%
- Zero-shot Test: 32.00%
- Fine-tuned Validation: 32.20%
- Fine-tuned Test: 31.80%

Limitations and Solutions

Computing BERTScore during every step of beam search increases inference time. Using a standard lexical overlap metric for intermediate pruning steps reduces computation time while preserving the final BERTScore evaluation for the completed sequence.

Correct Examples

	Fact	Stem	Choices	Gold	Predicted	Generation	BERTScores
1	deep sea animals live deep in the ocean	Frilled sharks and angler fish live far beneath the surface of the ocean, which is why they are known as	'Deep sea animals', 'fish', 'Long Sea Fish', 'Far Sea Animals'	0	0	deep sea creatures	[0.799683277027027, 0.7519120033725827, 0.7797675727146814, 0.7664145872567483]
2	Matter in the liquid phase has definite volume	Gas can fill any container it is given, and liquid	'is standard weight and size', 'is the opposite of variable', 'only needs a few', 'uses what it needs'	3	3	vapor	[0.7064760536804613, 0.6877207184619324, 0.7049006288435258, 0.7236700459866221]
3	migration is an instinctive behavior	When birds migrate south for the winter, they do it because	'they are genetically called to', 'their children ask for them to', 'it is important to their happiness', 'they decide to each year'	0	0	instinctive behavior is lacking	[0.8650879043910347, 0.8537702835235378, 0.85637274886557, 0.8480713311315337]
4	the amount of daylight is greatest in the summer	If a person enjoys long, sunny days their favorite month would be	'October', 'July', 'April', 'March'	1	1	June	[0.98291015625, 0.986328125, 0.9814453125, 0.98046875]

5	An example of hunting is an otter cracking open clams with a rock	An animal can hunt by cracking open a	'claw', 'house', 'shell', 'bone'	2	2	shell	[0.93994140625, 0.88427734375, 1.0, 0.955078125]
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Incorrect Examples

	Fact	Stem	Choices	Gold	Predicted	Generation	BERTScores
1	a compass is a kind of tool for determining direction by pointing north	If a person walks in the opposite direction of a compass arrow they are walking	'west', 'north', 'east', 'south'	3	0	in opposite direction	[0.8269184894304169, 0.812594609026746, 0.8254033901487398, 0.8103342622278557]
2	as an object moves , the kinetic energy of that object will increase	An example of lots kinetic energy would be	'Drinking a cold glass of water', 'A snail moving across the sidewalk', 'sitting without moving anywhere', 'An aircraft taking a trip'	3	2	a roller coaster	[0.73662974357919, 0.8089588638406231, 0.8505207026770355, 0.7656312181630054]
3	a single-cell organism cannot specialize	Which organism cannot specialize?	'mammal', 'plant', 'fish', 'protozoa'	3	0	moles	[0.9138727142285105, 0.7797888835877863, 0.7944279948702474, 0.8365501668733586]

4	a greenhouse is used to protect plants from the cold	A person can grow cabbage in January with the help of what product?	['Green house', 'Parliment', 'Congress', 'White house']	0	2	greenhouse	[0.9193586530801912, 0.9065465973344103, 0.9431885271092133, 0.9139779897836539]
5	an amphibian is cold-blooded	A frog, in winter, will burrow itself into soft mud, until it freezes, then in the spring	'it melts, because it is warm-blooded', 'it unfreezes, because it is cold-blooded', 'it remains frozen until it dies', 'it dies, because it is warm-blooded'	1	3	, it dies	[[0.8432534332758125, 0.8258226331577407, 0.8401773072175428, 0.9015821936175918]

Thoughts

The guided beam search produces identical or highly similar outputs to the vanilla approach.

The composite pruning score fails to alter the generation trajectory enough to correct the errors seen in Task 3. Relying on BERTScore for both intermediate pruning and final choice selection reinforces the same failure modes.